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Health Care Journeys of Veterans With Gulf War Illness

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Background: There is an acknowledged need to improve care for patients with persistent physical symptoms. Veterans who served in the 1990–91 Gulf War are a subpopulation of U.S. military Veterans who have been struggling with persistent physical symptoms for decades. The current study sought to characterize Veterans' historic path through the health care system and current experience of care to identify opportunities to improve care.

Methods: Analysis of interviews conducted with 31 Veterans who met criteria for Gulf War Illness (GWI) was conducted to understand Veterans' health care journeys, from symptom onset to the present.

Results: Early in their journey, Veterans felt uncertain about the nature of their condition and how to explain it to clinicians. Veterans described a cycle of referrals to specialists to pursue individual symptoms and subsequent return to primary care with few actionable findings. During this cycle, Veterans often felt dismissed or invalidated by clinicians. Over time, most Veterans felt care became increasingly fragmented, with multiple clinicians caring for them without a plan to manage GWI and little acknowledgement of GWI as a discrete illness. Further in their journey, some Veterans were referred to tertiary centers where they encountered a more holistic approach.

Conclusions: Findings point to the need to shift care for Veterans with GWI, and similar conditions, away from overly focusing on individual symptoms. Instead, primary care clinicians need training and support, potentially from tertiary care experts, to develop and implement holistic care plans that recognize GWI as a complex chronic condition.

Key Words: environmental exposures, healthcare journey, Gulf War Illness, Veteran, persistent physical symptoms, pain, military exposure, qualitative, trust

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Gulf War Illness (GWI), a symptom-based condition associated with military exposures,[1–3] impacts ~30% of Veterans who served in the Persian Gulf in 1990–1991.[4] GWI is characterized by disabling persistent physical symptoms including chronic fatigue, widespread pain, and gastrointestinal symptoms.[1,5] GWI is like other symptom-based conditions such as myalgic encephalomyelitis/chronic fatigue syndrome and fibromyalgia, which are poorly understood and difficult to diagnose and treat.[6–9]

There is an acknowledged need to improve health services for Veterans living with GWI and other symptom-based conditions. Veterans living with GWI have long fought to have the condition recognized by the medical community and describe feeling betrayed by the US Government's response to their health concerns and needs.[8,10] These negative experiences have been attributed to several factors, including low clinician knowledge,[11]

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under-recognition of GWI in practice settings,[12] disagreements about the nature of GWI between Veterans and clinicians,[13–15] and stigma associated with contested or poorly understood illness.[16,17] Studies on GWI and other symptom-based conditions highlight the need to improve health services and have primarily identify clinician-level and individual-level barriers and facilitators to high-quality care.[18]

System-level barriers (eg, administrative burdens and resource allocation) are some of the strongest contributors to quality of care among other populations.[19,20] System-level factors are understudied for symptom-based conditions; prior research has found discrete barriers to quality care for these conditions, including health care fragmentation,[9] the absence of clinical practice guidelines (CPGs), and time-limited appointments.[18] For GWI specifically, system-level factors impacting care have included a lack of clinical case definition for GWI[2] and a CPG that recognizes GWI as a discrete condition not being developed until 2014.[21]

While identification of discrete system-level barriers impacting care is important, what is not well described is the trajectory through the health care system from the patients' perspective, termed "health care journeys."[22–24] How patients move through the health care system can identify both barriers to care and potential points of intervention.[25,26] We are aware of only one study of health care journeys of patients with symptom-based conditions, specifically fibromyalgia, which highlighted difficulties with access to specialty care.[27] The current study sought to characterize the historic path of Veterans with GWI through the health care system and the current experience of care to identify points of intervention at the system-level. Analysis of health care journeys, an accepted qualitative research method, was conducted with a purposeful sample of Veterans living with GWI (N = 31)[26] to help improve health services for Veterans with persistent symptom-based conditions.[25]

The mapping and analysis of Gulf War Veterans' health care journeys offers a way to examine variability in system-level barriers among a subpopulation that served during the same time-period and began seeking care for unexplained symptoms shortly thereafter. Veterans in the study also received health care from Veterans Health Administration (VHA), a single-payer health care system, which eliminates some financial barriers and operates under enterprise policies. Understanding system-level barriers and facilitators among a population with lower historical variability and fewer financial burdens may help identify unique points of interventions that may benefit all patients with symptom-based conditions.[27]

METHODS

Participants

Participants were recruited as part of a larger mixed-methods study of the role of patient-clinician concordance in health care experiences of Veterans living with GWI (Phillips et al,[28] for details about the parent study). Veterans could participate in the parent study if they either had an upcoming appointment at the VA's tertiary center for GWI, the War Related Illness and Injury Study Center (WRIISC),[29] which provides an intensive 2–3-day long evaluation with a multidisciplinary team for Veterans referred by their primary care clinicians, or a primary care clinic located in the same VA medical centers as the WRIISC sites (ie, Northern New Jersey, Northern California, and the District of Columbia). Veterans had to be deployed to the Persian Gulf in 1990–1991 and meet the Kansas definition for GWI,[1] which is recommended by the National Academy of Medicine. The Kansas definition of GWI encompasses at least one moderate to severe chronic symptom in 3 or more of the following categories: fatigue/sleep, pain, neurologic/cognitive/mood, gastrointestinal, respiratory, and skin. Patients do not meet Kansas GWI criteria if they have another diagnosed medical condition that better explains their symptoms (eg, multiple sclerosis, rheumatoid arthritis, and congestive heart failure).[1]

Upon enrollment in the parent study, Veterans were administered several standardized measures, one of which focused on perceptions of concordance with their medical provider regarding their symptoms.[14,30] We sought to conduct interviews with a sample of Veterans with high and low concordance scores. On a rolling basis, Veterans (N = 31) were invited and agreed to participate in qualitative interviews until we achieved an equal mix of Veterans reporting high and low concordance with their clinician about their GWI (Table 1).[14] One Veteran who was contacted to take part in the qualitative interview did not agree to be interviewed.

Procedures

Interviews lasted ~60-minutes and were conducted from February 2017–July 2018. The interview guide (Supplement 1, Supplemental Digital Content 1, http://links.lww.com/MLR/D95) was developed by the research team and guided by research questions regarding symptoms and impact of symptoms on daily life, illness beliefs, perceived concordance between the patient and their clinician about GWI and factors associated with perceived concordance. Interviews then explored Veterans' health care experiences related to seeking treatment for GWI symptoms. Participants were compensated for their time. All participants provided informed consent, and all study procedures were approved by the VA Central Institutional Review Board.

Data Analysis

Three researchers reviewed a sample of interview transcripts and then met to develop a draft codebook. They applied this initial codebook to a second set of transcripts, refined the codes, and developed new codes. The team continued to meet to reach consensus on the final codebook, which entailed reviewing and coding additional transcripts to build consensus on code definitions and application. The remaining transcripts were dis-

TABLE 1. Demographic Characteristics(n) of Respondents

Sex	
Male	29
Female	2
Race	
African American/Black	9
Hispanic	5
Native Hawaiian	1
White	16
Care location recruited from	
Primary care	12
Tertiary referral/WRIISC	19
Concordance	
Low	16 (9 WRIISC)
High	15 (10 WRIISC)
Mean age (Age range)	52 years (45–60 years)

(N = 31).

tributed to team members to code independently. The team met routinely to discuss and resolve coding challenges and to further refine code definitions. Interview transcripts were uploaded to NVivo, and codes were applied electronically. The first author completed a descriptive timeline for each participant guided by thematic codes in the codebook (eg, impact of illness on life) and new codes that emerged from the timeline (eg, the primary-specialty care cycle).[24,26,31] These timelines were then reviewed and summarized for the current analysis in consultation with coauthors. The timelines, thematic codes, and subsequent construction of health care journeys used a framework matrix method to organize the data.[32]

RESULTS

Participants

Table 1 shows the demographic characteristics of the sample. Most participants identified as male (94%). Half of the sample (51%) identified as White, 29% identified as African American/Black, 15% as Hispanic, and one Veteran identified as Native Hawaiian. The mean age of the participants was 52 years (range: 45–60 y).

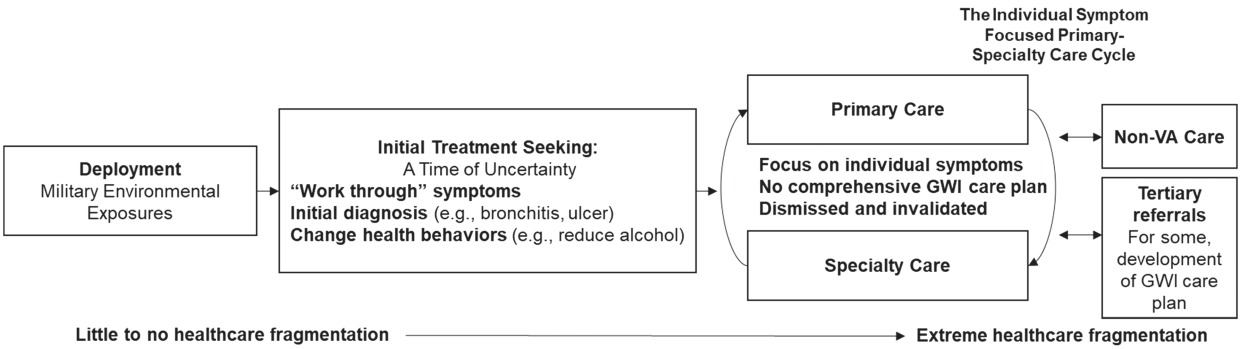
Overview of the Health Care Journey

Figure 1 provides an overview of these journeys as well as 2 examples of themes and illustrative quotes. Generally, journeys started with deployment to the Persian Gulf region in 1990–1991 and an initial presentation of persistent physical symptoms during deployment or shortly after the Veteran's return. When first seeking care for symptoms, Veterans felt uncertain about their health and how to present concerns to clinicians. Symptoms were often attributed to health behaviors (eg, not sleeping and drinking alcohol) or temporary illness (eg, bronchitis, ulcers). Despite initial explanations and associated treatment, symptoms persisted. As symptoms manifested and became chronic and increasingly disabling, Veterans found themselves in a primary–specialty care cycle, where primary care clinicians referred them to specialists to evaluate individual symptoms (eg, chronic diarrhea to gastroenterology) both in VA and non-VA care settings. When tests came back "normal" or specialists could find no underlying cause for the Veteran's symptoms, the Veteran returned to primary care. Many felt this time was marked by the clinicians' search to explain individual symptoms as opposed to a recognition of the need to manage a distinct complex illness. Veterans also felt a lack of a cohesive care plan. Over time, some were referred to tertiary centers, like the Mayo Clinic or WRIISC. Veterans found recommendations from the WRIISC focused on managing GWI as a complex chronic condition and often included recommendations for integrated care (ie, low inflammation diet, acupuncture). However, following through on referrals for these services was challenging. For many, as symptoms persisted, care became increasingly fragmented.

Initial Treatment Seeking: A Time of Uncertainty

When initially seeking care for what we now call GWI, Veterans felt uncertain about their condition and how to explain what was happening to their bodies. This uncertainty extended to the cause of symptoms, initial treatments, and whether symptoms were part of an overarching condition or syndrome. During deployment, Veterans reported exposures including airborne hazards (eg, burning oil wells), vaccinations, prophylactic medication use, and physical and psychological stress injuries. While Veterans were sure they were exposed to something, most were unsure how these exposures related to their symptoms. Most were also relatively young (early-to-mid 20s) and generally inexperienced with managing their health and seeking health care. As one Veteran noted, "I wasn't able to articulate it very well…which didn't help them in their attempts to try to figure out what's going on and come up with a treatment plan" (Participant 8). Veterans stated they often first discounted their own condition, referencing their ability to "work through" (Participants 9 and 28) symptoms or accept initial diagnoses of acute or time-limited conditions (eg, "they thought maybe it was an ulcer" Participant 21). Veterans noticed that while symptoms temporarily abated when they quit smoking or heavily drinking alcohol, symptoms eventually returned.

Veterans became increasingly concerned about their deteriorating health condition and prognosis. Veterans reported difficulty describing symptoms of a condition that the medical community had not yet defined, "How can you have a syndrome or illness and have symptoms for it, but technically there is nothing called that?" (Participant 19). As Participant 20 noted, "in September of 1991 no one had any idea that there was a Gulf War Syndrome or Illness." However, early on in their illness trajectory, Veterans reported that in talking with others, "a lot of them had all these different symptoms" (Participant 21). Others saw news stories about GWI-like symptoms and, as one Veteran stated, "you don't want to think about it. You don't want to be another victim" (Participant 14).



	Deployment	Initial Treatment Seeking	Primary-Specialty Care Cycle	Extreme Healthcare Fragmentation
Veteran 10	"Over there, I was…where the planes would go on their bombing missions, come back, our guys in one area would check for chemicals. Like nerve gas and stuff like that."	"I didn't put two-and-two together until I started getting this Desert Storm Newsletter. And I really never read them because I just thought I just, I'm getting older. This stuff happens…And I started reading it, and I got looking at what everybody had been going through and the symptoms, and out of the 10, I think I had seven of them so far."	"As soon as we showed her [primary care clinician] that paper [on Gulf War Illness], I think she started researching it. And she got it, but she didn't know where to go with it. Because, okay, he's got all these symptoms of what was happening. But what do I do?"	"I ended up in the [Name] Hospital for four days, and their conclusion was we don't know what's going on …And the same thing happened when I went to [Name] Hospital…they sent me down to the [tertiary center] … That doctor sent me to an internal medicine doctor. That internal medicine doctor wanted me to go see the pulmonologist there. So, she sent me to another internal medicine doctor."
Veteran 14	"You know, initially or in the 90's, I had no idea that I had been affected by everything I went through out there…I've forgotten so much…I was able to reflect back on everything I went through and how the environment and exposures and everything affected me. I could reflect on how sick I was out there from anything from eating food to getting bitten and inhaling all the stuff we inhaled out there."	"As the years went by, you know, when I would recognize certain symptoms and, and, and go through certain life-changing events, or if I would get sick, I would just blame it on something else or assume it was something else, or just ignore it all together. Until several years ago, everything just came to a halt."	"I thought as though the doctors were not listening to me or just unwilling to give me the necessary time, the symptoms were there, but they couldn't figure out why. They would run certain exams and testing, but the funny thing is, as they would sit there and tell me, 'Well I don't think that's the problem' because I was reading a lot of the medical information regarding my symptoms, and of course I'm not a doctor…many times I would leave there feeling worse than what I had already felt..."	"I felt good when I got back [from the tertiary center], but then I, honestly, I kind of feel like since I got back, I've been going backwards. I anticipated being contacted and having these appointments set up immediately for the recommendations, but it didn't happen. And at this point, I don't even know what to do, but I think I'm going to have to contact the VA and make an appointment with the new doctor because I haven't heard from them."

FIGURE 1. Health care journeys of Veterans living with Gulf War Illness with examples of Veterans' narratives and corresponding themes.

The Individual Symptom Focused Primary-Specialty Care Cycle

As Veterans' symptoms became worse, clinicians referred Veterans to specialists to evaluate individual symptoms. This became a cycle where Veterans were referred by primary care to various specialists and returned to primary care without answers or solid treatment recommendations. For many Veterans, this cycle continues into the present day. Veterans described specialty care referrals as being driven by a search for the cause of individual or clusters of symptoms without a concrete or actionable finding. The lack of biomarkers or measures of GWI appeared to challenge clinicians. As Participant 10 noted, "every [test] she sent me to, I passed…And that confused the hell out of her, all the doctors, and everybody else." Veterans repeatedly shared that test results came back "negative" or within the normal range, which "leads to nowhere" (Participant 3), yet their symptoms persisted.

Veterans also described symptoms not being seen as part of a complex chronic illness. Participants described conventional defensive medicine approaches to their care, which focused on referrals and testing by symptom, thus leading to a fragmented approach and perspective. Veterans perceived clinicians as denying that symptoms were caused by Veterans' military service, and not believing in GWI as a discrete condition even in the present day—"no one really talks about Gulf War Illness" (Participant 18). Some described having positive relationships with their primary care providers despite a persistent lack of knowledge about GWI. As Participant 1 noted, their clinician is "not too familiar with the Gulf War Illness, but she has gotten into it a little from my understanding. She's a very good doctor. She's very caring and understanding."

Dismissed and Invalidated

Throughout this individual symptom focused primary-specialty care cycle, Veterans often described feeling dismissed and invalidated. Recommended treatments discussed in primary and specialty care were inconsistent with Veterans' lived experience of GWI. For example, while many Veterans felt recommendations related to changes in health behaviors (eg, diet) were useful, others noted that things like exercise and dietary changes were challenging or even impossible due to pain and fatigue. As Participant 3 states, "I don't think they have a grasp or a real understanding of the reality of the symptoms…like eating healthy, well that sounds good but if you don't

have someone that's going to the store for you, cooking for you, cleaning for you…their suggestions are great, but they don't match with my reality."

Particularly in the initial years of their health care journey, many Veterans felt clinicians invalidated the severity of their GWI and its impact on their lives. Veterans perceived GWI as disabling but felt their clinicians did not share this view. For example, Veterans felt clinicians did not understand their level of functioning before GWI became disabling. Comments such as, "I was very athletic and very outgoing. Now…I'm barely active…I totally stay in the house. I'm like a hermit…" (Participant 4), were common.

Non-VA Care

Many Veterans appeared to move between VA and non-VA care, specifically specialists, with regularity. Here, non-VA care often introduced recommendations for integrative approaches that many found helpful (eg, acupuncture, specialty diets). However, it also involved recommendations for more invasive treatments, some with little evidence of efficacy or relevance (eg, chelation). One Veteran felt that they were encouraged to seek non-VA or private health care by VA clinicians as an alternative to continuing with VA care, sometimes undermining trust in the VA. Participant 31 did not tell their non-VA clinician about their time in the Persian Gulf because "I feel as though it's not his issue." A few Veterans described negative experiences with non-VA care. For example, one Veteran who sought care from an academic medical center stated, "[the rheumatologist] told me that I wasted $10,000 of the taxpayers' [dollars], there was nothing wrong with me" (Participant 27).

Fragmentation of Care and Lack of a GWI Management Care Plan

Over time, Veterans found themselves with extremely fragmented care. This was caused by receiving treatment from multiple clinicians, including in private sector health care, academic medical centers, and tertiary clinics. Veterans shared that they tried a variety of tests, procedures, and treatments over many years and met with numerous clinicians. As Participant 24 states, "I have had 10 different dermatologists…I've been to countless doctors, both civilian and VA…" A second Veteran states, "I'd reached the point where I was going to…four or five different doctors for four or five different things…not really aware of what the other person was doing or how that interacted with what the other person was doing" (Participant 24). This culminated in Veterans feeling frustrated and lost, with multiple clinicians across multiple systems managing individual symptoms with no care plan for GWI as a distinct condition. In contrast, Veterans who reported they had a "plan of attack" (Participant 2) tended to feel more satisfied with their care.

Tertiary Centers

Several Veterans indicated that they would like to be referred to a clinician who specialized in or was highly knowledgeable of military exposures in general, and GWI specifically. One Veteran stated, "I feel as though if it was too much for a physician, then maybe they should just refer me to someone else that can treat Gulf War Veterans" (Participant 14). About half of the sample had the experience of receiving consultative care from tertiary centers (eg, VA's WRIISC or the Mayo Clinic), particularly the WRIISC. These consultations were often the first time Veterans' military service or exposures were explicitly discussed in a clinical context. Similarly, tertiary center clinicians were often the first to view their GWI holistically as a complex chronic condition that needed to be managed, as opposed to focusing on individual symptoms. Veterans reported they received an evaluation explicitly for their GWI at tertiary centers, stating that these centers were "holistic" (Participant 24) and gave a "roadmap" (Participant 3) for managing their GWI.

Like non-VA care, management recommendations from tertiary centers often included integrative practices (eg, acupuncture, specialty diets). However, the incorporation of recommendations from GWI specialists into primary care was a challenge. Veterans noted that bringing these recommendations back from tertiary centers required additional support to both discuss recommendations with their primary care clinician and to implement them (eg, exercise with chronic pain, improving diet with limited access to grocery stores).

DISCUSSION

The health care journeys of Veterans with GWI highlight how health care systems are often ill-prepared to care for symptom-based conditions, especially those related to military exposures. Early in their journey, Veterans with GWI felt uncertain about the nature of their condition and how to explain their symptoms. Veterans described a cycle of referrals to chase down individual symptoms and a subsequent return to primary care with few actionable findings; for many Veterans, this cycle persisted in the present day. Veterans felt their care became increasingly fragmented, without a clear comprehensive plan for how to manage symptoms. Some Veterans were referred to tertiary centers where they encountered a more holistic approach.

A potential system-level driver of these health care journeys is the poor fit of the biomedical model of care with conditions characterized by uncertainty, including GWI and military exposure concerns. The biomedical model suggests "…the body can be seen as a machine. Its parts all work together and when, broken, can be fixed or replaced, part-by-part."[33] This piecemeal approach has created challenges for the Veterans in our study as well as other symptom-based conditions, like fibromyalgia.[9] This is because the biomedical model encourages clinicians to focus on identifying the "malfunctioning part,"[26] which can lead to multiple referrals and unnecessary assessments as opposed to recognition of a discrete chronic illness requiring evidence-based management. A holistic approach, that places the patient's goals for functioning at

the center, may help clinicians shift from this model and improve the experience of care for GWI.[34,35] Veterans in our study stated that more than chasing individual symptoms, they wanted a conversation about GWI and military exposures and a "road map" to manage their symptoms. Part of this struggle is that there is no "cure" for GWI. Care plans to manage symptoms can be informed by the Clinical Practice Guideline for GWI,[21] which is consistent with a holistic approach to health care.[34]

Successfully moving clinicians from a "part-by-part" approach[33] to helping Veterans to manage the symptoms of GWI requires system-level support.[36,37] Training and education for clinicians on GWI should include evidence-based strategies to manage symptoms of GWI, found in the Clinical Practice Guideline (CPG) for GWI, and effective communication skills, which includes validating the Veteran's experience with GWI, developing a shared understanding of the condition, and using shared decision making to establish a concrete care plan.[38,39]

In addition to training, addressing the extreme care fragmentation perceived by Veterans in this study also requires system-level initiatives. Care fragmentation is a major theme reflected in the experiences of care for persistent physical symptoms and symptom-based conditions[9] and is often associated with excessive, unnecessary, and potentially harmful interventions[40–42] in both primary and specialty care.[43] Concrete interventions have been suggested to mitigate care fragmentation and improve advocacy, including case management,[44] health coaching,[35,45,46] and interdisciplinary collaboration.[47] These interventions can also take some of the burden of these complex conditions off primary care. Increasing the availability of support to Veterans and clinicians from specialists who provide consultation and care for the most complex cases may additionally improve care fragmentation.

Veterans who had the experience of an intensive consultation with a specialized tertiary team reported the experience was positive. Whereas primary care and core specialty care clinicians (eg, gastroenterology and dermatology) had not necessarily recognized their symptoms as being related, clinicians within the tertiary center who specialized in GWI did. We aimed to recruit ~50% of our sample from the WRIISC and may be able to learn more about providing care from tertiary centers' best practices and translate these back into primary and core specialty care settings. However, while tertiary centers were a place where Veterans felt their GWI was addressed holistically as a condition related to military exposures, integrating recommendations from the center clinicians back to primary care was a challenge. This calls for a greater understanding of how to integrate wholistic consultations from tertiary centers and into routine primary and specialty care.

Limitations

This study offers insight but should not be interpreted as being representative of the experiences of all Veterans with GWI. The goal of this study was to gain an understanding and insight into the unique experiences of participants.[48] This study was not without limitations. It does not capture the experiences of Veterans with GWI who never use VA health care and includes a high proportion of Veterans referred to the WRIISC, which may suggest a high representation of those with severe symptoms (54). In addition, we did not distinctly capture the experience of Veterans with symptom-based conditions who developed symptoms from other conflicts, who did not serve in combat, or non-Veterans. Finally, asking participants to recall events from the distant past and about a broad subject (eg, health care seeking or concordance) is subject to recall bias.

CONCLUSION

Veterans' health care journeys related to GWI suggest that to effectively support those living with conditions like GWI requires system-level interventions to improve Veterans' experience of care for GWI. Findings point to the need to shift care for Veterans with GWI, and similar conditions, away from overly focusing on individual symptoms. Instead, primary care providers need training and support, potentially from tertiary centers, to help develop and implement holistic care plans that recognize GWI as a discrete and complex chronic condition.

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